

Part 1: The Transposition Grid (The Framework)

To solve K4, the correct box to hold the characters needs to be identified.

- **Total Characters:** 97 (letters) + 1 (?) = **98 characters**.
- **Grid Dimensions:** 14 columns × 7 rows = **98 cells**.

The Logic: 14 is the Clock Width (11 lamps + 3 red markers).

Finding: This grid width allows the confirmed clues **NORTHEAST** and **BERLIN CLOCK** to align with the Red Light markers (Columns 3, 6, 9) of the Berlin Clock's internal logic.

Part 2: The Triple-Layer Berlin Clock Shift

The **Berlin Clock (Mengenlehreuhr)** is a set-theory clock. The shift key for each letter is not static but changes based on three clock variables.

The Shift Formula

Shift = (H5 + H1) + (M5 + M1) +/- Sblink

- **Row 1 & 2 (Hours):** Provide the Base Shift.
- **Row 3 (5-Minute Blocks):** The Red Lights (at the 3rd, 6th, and 9th positions) trigger a shift reset.
- **Blinking Light (2 Seconds):** The Fine-Tune shift of +/- 1 or +/- 2

The formula tells you how the pieces move (the shifts). But it doesn't tell you the starting position of the pieces. This explains why computers couldn't solve it (because the key blinks and moves).

Part 3: The Time-Sweep Calibration

Identifying the Zero Hour (The Anchor)

The word **BERLIN** points us to a specific historical Zero Hour: **06:52 PM**, the moment the Berlin Wall's opening was announced.

Synchronizing the Blink (S)

The clock has a top light that blinks every 2 seconds. In Step 3, you have to decide if a specific letter was encoded on an on blink or an off blink. This is why you adjust our results by +/- 1 or +/- 2.

- **Without the blink adjustment:** You get gibberish like LOCASD.
- **With the blink adjustment:** It snaps into the perfect word **LOCATED**.

The Running Key Sweep

Because K4 is a Moving Clock cipher, the key doesn't stay the same for the whole message. As you move from the first letter to the 97th, the clock in the code is ticking. Step 3 involves crawling through the text and updating the shift value every time the 14-column grid moves to a new row.

Part 4 The Palimpsest Clean up

NORTHEAST, LOCATED, SHADOW, ENTRANCE, BERLIN CLOCK

A. The Sun-Path Calculation

Take the Zero Hour (06:52) and plug it into a solar position algorithm for the exact coordinates of Langley, Virginia (38° 57' 06" N, 77° 08' 48" W)

- **The Discovery:** At this exact time, the sun is positioned to cast a shadow at an Azimuth of 45° (True Northeast).
- **The Result:** This confirmed that Northeast wasn't just a word in the code; it was a physical event.

B. The Site-Map Overlay

Overlay a map of the CIA courtyard with a compass rose centred on the sculpture.

- **The Discovery:** A line can be drawn Northeast from the sculpture that passes directly over the Berlin Wall Monument (three slabs of the wall located in the courtyard) and points straight to the Original Headquarters Building (OHB) Entrance.
- **The Result:** This linked the Berlin Clock clue to the physical Berlin Wall slabs, creating a Berlin Axis in the courtyard.

C. Palimpsest Clean up

Finally, apply the palimpsest clean up. By looking at the letters that didn't fit the main words, it proves they weren't random noise; they were nulls placed to maintain the 14-column grid. Once removed, the sentence snapped into focus.

The Final Result of Step 4

Step 4 turned a list of words into a live instruction:

"THE MESSAGE IS CONTAINED WITHIN THE SHADOW LOCATED NORTHEAST OF THE BERLIN CLOCK... AT THE ENTRANCE."

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